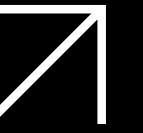
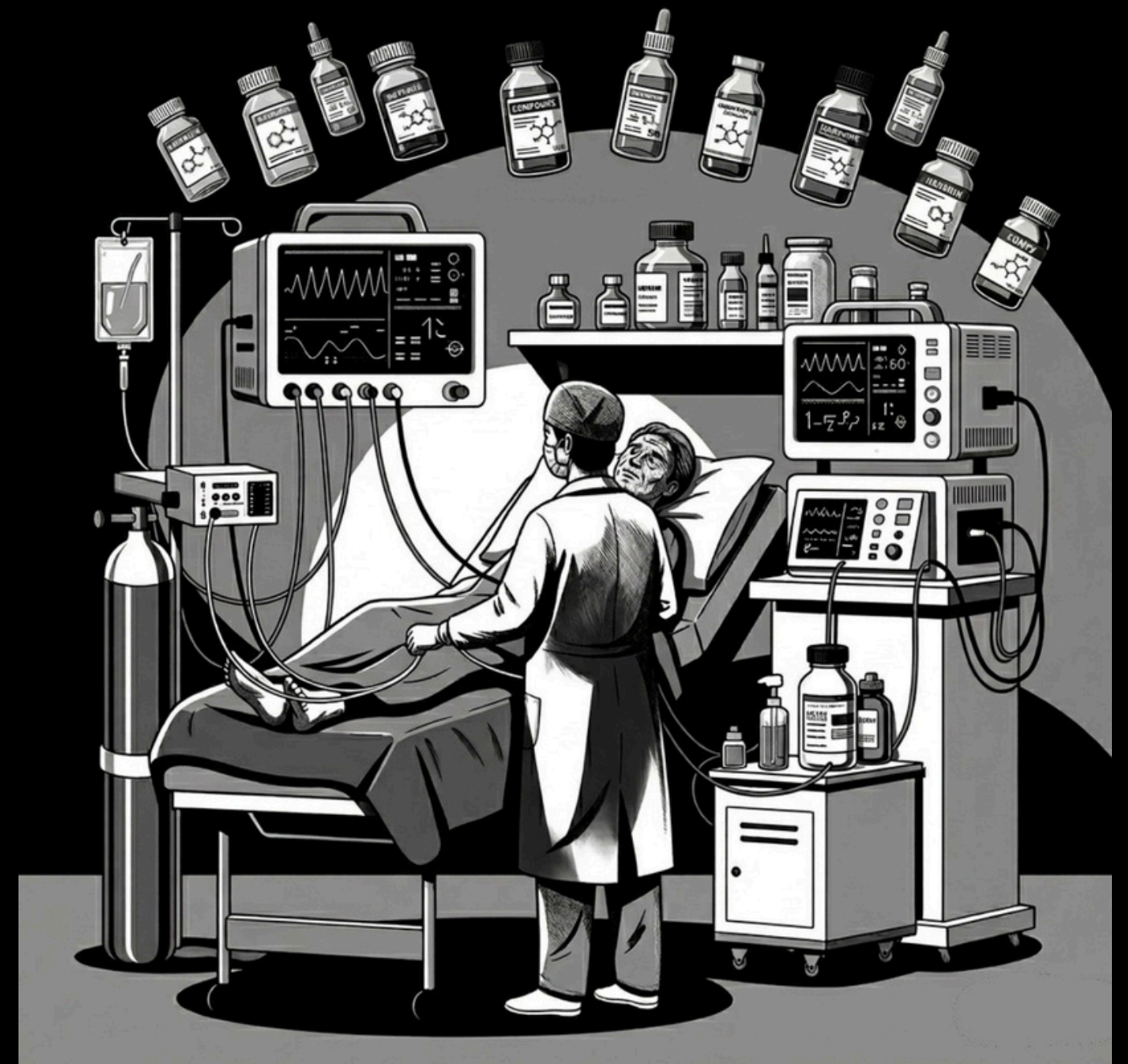


ICU Risk Prediction Technical Updates

Alain Areeba Siddiqui · Mariah Noelle Cornelio · Darlene Almazan Eligado

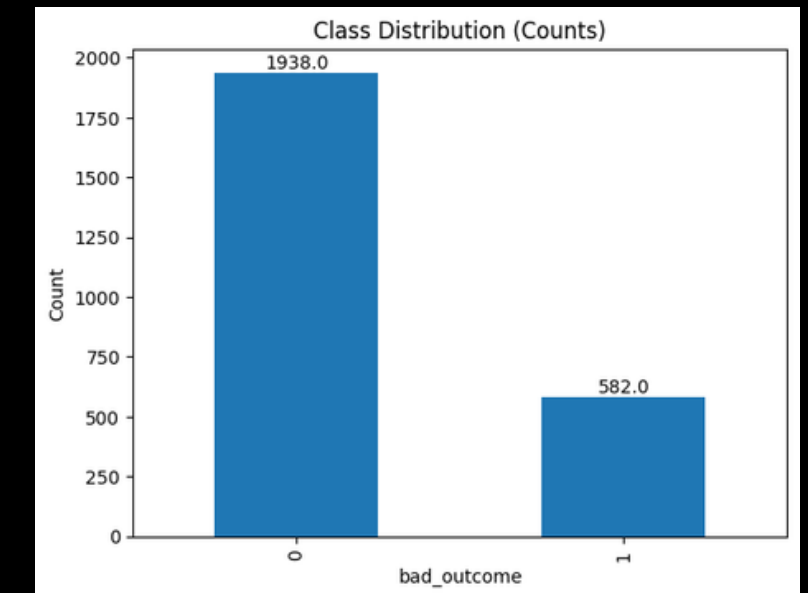
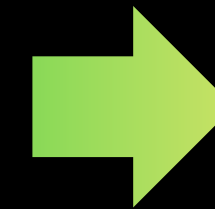
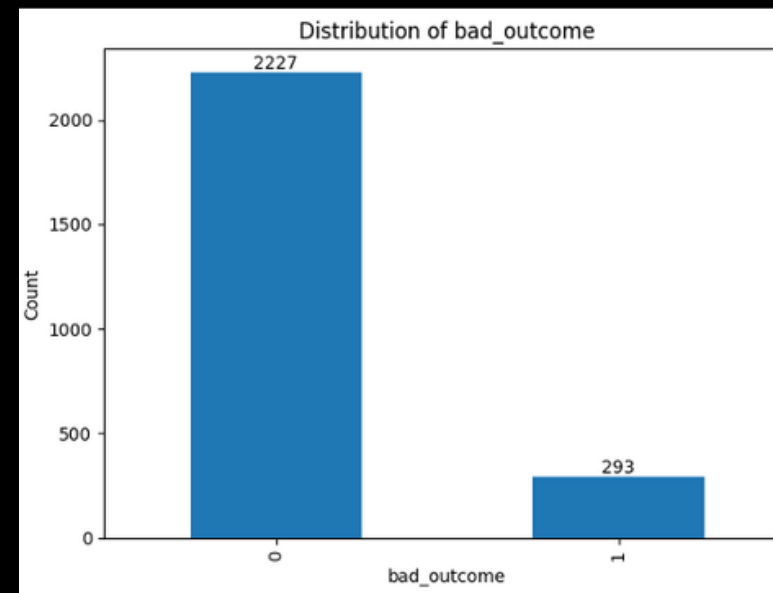


What Changed +

1. Remerging + Starting from Scratch

- Old DataFrame: 55 features
- New DataFrame: 87 features
 - Patient past history
 - Allergies
 - Diagnosis priority

+ 32 features



2. Added to bad_outcome definition

- Bad Outcome Risk: Death and Readmission
 - Status = "Expired"
 - Readmission = 1
- unitvisitcount > 1

+ 289 Bad Outcomes

Baseline Comparisons

Old Dataset

Logistic Regression Baseline Model			
	Precision	Recall	F1
0	0.89	0.99	0.94
1	0.64	0.15	0.24
Macro Avg	0.77	0.57	0.59

ROC-AUC: 0.78

PR-AUC: 0.39

New Dataset

Logistic Regression Baseline Model			
	Precision	Recall	F1
0	0.85	0.95	0.90
1	0.73	0.46	0.56
Macro Avg	0.79	0.70	0.73

ROC-AUC: 0.82

PR-AUC: 0.67

Improvements (NOT CLICKBAIT):

+ 14% Class 1 Precision

+ 206% Class 1 Recall

+ 133% Class 1 F1

+ 5.1% ROC-AUC

+ 71.8% PR-AUC

Final Models

Model 1: LR + CB + XGB + RF > LR Meta

Cost-Based Model			
	Precision	Recall	F1
0	0.84	0.97	0.90
1	0.79	0.41	0.54
Macro Avg	0.82	0.69	0.72

ROC-AUC: 0.83

PR-AUC: 0.68

Model 2: LR + CB + XGB + SVM + RF > LR Meta

Supporting Model			
	precision	recall	f1-score
0	0.914	0.825	0.867
1	0.558	0.741	0.637
Macro Avg	0.736	0.783	0.752

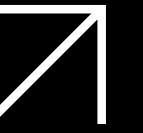
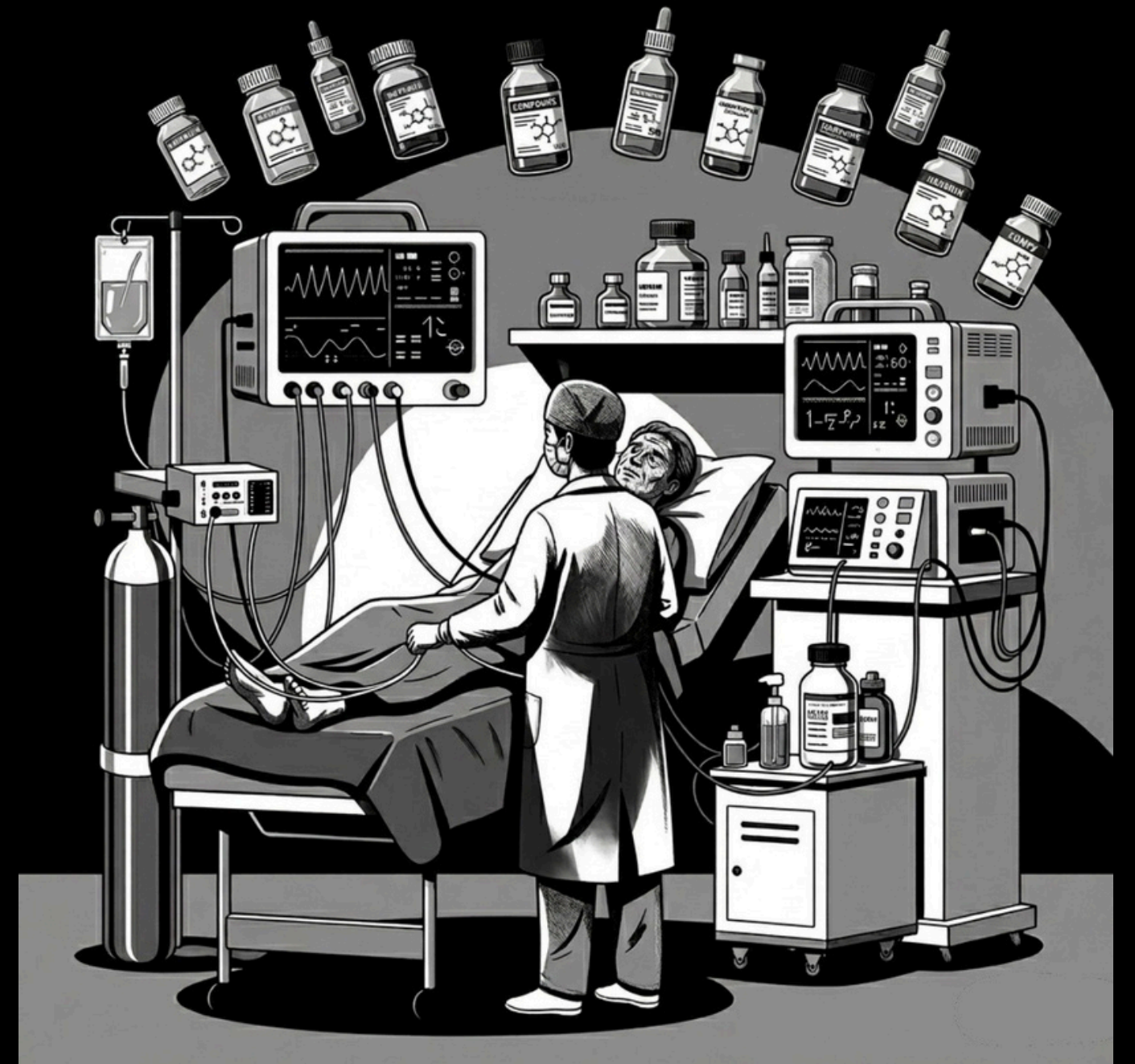
ROC-AUC: 0.85

PR-AUC: 0.71

2 different models = to catch as many critical patients as possible AND avoid unnecessary ICU alarms

ICU Risk Prediction

Alain Areeba Siddiqui · Mariah Noelle Cornelio · Darlene Almazan Eligado



Dr. Kyna Fong



Stanford



CEO and Co-Founder of Elation
Health, Health Economist, and
Practice Manager

- **Elation Health** is an EHR (electronic health record) company that focuses on efficient AI built for independent primary care and is the largest and most trusted clinical technology platform for primary care in the U.S
- "I built Elation to help clinicians deliver high-value, personalized care."
- Today, Elation's AI-powered EHR support 46,000 clinical users caring for over 23 million patients



Forbes

Goldman
Sachs

The Problem

Current clinical decision-making is often:

- Time-constrained
- Reliant on individual experience
- Not supported by unified predictive tools
- Limited interpretability when ML is used
 - Narrow trust in predictive models

Why It Matters

ICU risk prediction is critical because:

- Improves survival outcomes and preventable deaths
- ICU resources are limited and must be prioritized effectively
- Clinicians AND patients need tools that are trustworthy and interpretable

Stakeholders & Impact

Patients

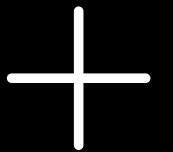
- Improved Personal Care
- Transparent Communication

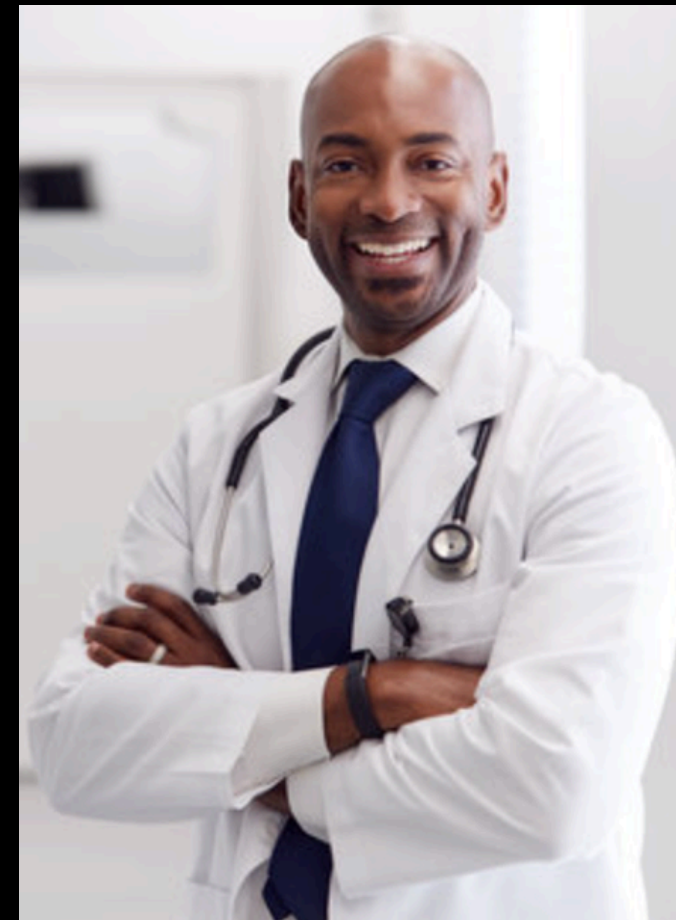
Healthcare Providers

- Augmented Decision-Making
- Reduced Cognitive Load = Less Burnout

ICU Facilities

- Efficient Resource Allocation
- Operational Quality Control





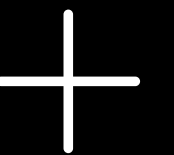
What Data Can and Cannot Tell Us

What It Can Tell Us:

1. Patient Risk Scores
2. Resource Planning
3. Clinical Patterns between patient vitals

What It Cannot Tell Us:

1. Causality v. Correlation
2. Response to treatment
3. Ethical Considerations



Key Insights

Primary Signal

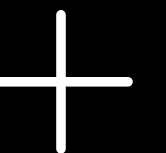
- Expanded features for stronger risk detection
- Redefined outcome captures real ICU risk

Model Behavior

- Prioritized recall & precision better for identification of critical patients
- Performance improved on imbalanced data

Decision Support

- Dual model agreement increases confidence
- Explainability (SHAP) aligns with clinical intuition



Recommendation

- **Deploy as Decision Support**
 - Not a replacement for clinicians
 - Integrated into ICU workflow
- **Operational Use**
 - Real time risk monitoring
 - Threshold base alerts
- **Trust & Adoption**
 - Dual model validation
 - Explainability for transparency

Impact

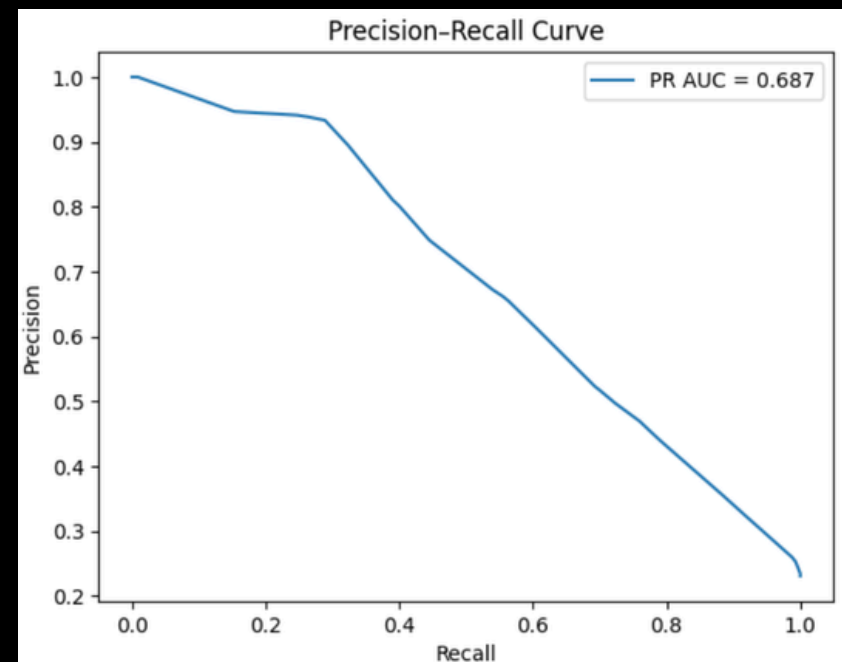
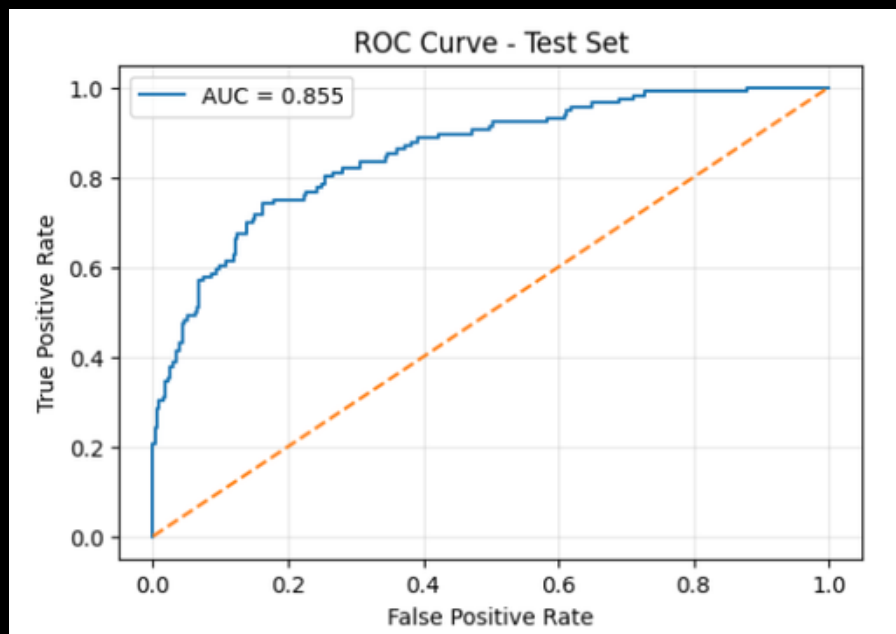
- **Clinical Impact**
 - Earlier risk detection
 - Reduced preventable deaths
- **Operational Impact**
 - Better ICU resource allocation
 - Reduced readmissions
- **Strategic Value**
 - Scalable across hospital systems
 - Supports data driven care decisions



Confidence & Uncertainty

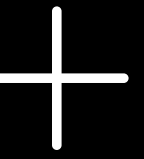
Validated Strengths

- Strong ROC & PR performance
- Improved detection of high-risk patients
- Explainability supports clinical trust



Uncertainty

- Limited dataset size
- Static 24-hour snapshot
- Sensitive to data quality



Risks & Consequences



Under Predictions (False Negatives)

- Miss high risk patients
- Delayed intervention



Over Predictions (False Positives)

- Unnecessary interventions
- Resource strain



Mitigation

- Human-in-the-loop validation
- Threshold tuning
- Continuous monitoring + retraining



Deployment



Thank you!

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